

Patient Profile

Name: Saubhik Bhaumik

Age: Not provided

Gender: Not provided

Summary of Results

All evaluated laboratory parameters fall within their respective reference ranges. The predictive model has classified the overall health profile as **Normal**, indicating no evidence of abnormal patterns requiring immediate clinical intervention.

Detailed Test Explanations

- **Serum Bilirubin (Total):** 0.9 mg/dL (Normal: 0–12 mg/dL). Total bilirubin reflects hepatic excretion; values within range suggest preserved hepatic clearance and no overt cholestasis or hemolysis.
- **Serum Bilirubin (Direct):** 0.2 mg/dL (Normal: 0–3 mg/dL). Direct (conjugated) bilirubin elevation can signal biliary obstruction or hepatocellular dysfunction; the low value confirms absence of conjugated hyperbilirubinemia.
- **Serum Bilirubin (Indirect):** 0.7 mg/dL (Normal: 0–2 mg/dL). Indirect bilirubin rises primarily with hemolysis or impaired conjugation; the value lies within normal limits, indicating stable erythrocyte turnover and intact hepatocyte conjugation activity.
- **SGPT/ALT:** 36 U/L (Normal: 13–40 U/L). Alanine aminotransferase is a marker of hepatocellular injury; the measurement is within the reference range, suggesting no current hepatocellular damage.
- **SGOT/AST:** 32 U/L (Normal: 0–37 U/L). Aspartate aminotransferase reflects liver, cardiac, and muscular injury; the result is normal.
- **Serum Alkaline Phosphatase:** Not reported. Absence of data precludes interpretation.
- **Serum Protein:** 7.2 g/dL (Normal: 6.4–8.3 g/dL). Total protein encompasses all circulating plasma proteins; the value indicates adequate protein synthesis.
- **Serum Albumin:** 4.7 g/dL (Normal: 3.5–5.2 g/dL). Albumin is a marker of hepatic synthetic function and nutritional status; the level is within normality.
- **Globulin:** 2.5 g/dL (Normal: 1.8–3.6 g/dL). Globulins include immunoglobulins and transport proteins; the measurement is normal.
- **Albumin/Globulin Ratio:** 1.88 (Normal: 1.1–2.4). This ratio reflects balance between albumin and globulin; the value falls comfortably within the reference interval.

Overall Interpretation

The laboratory panel demonstrates complete biochemical health without evidence of hepatic compromise, cholestasis, hemolysis, or protein synthesis disorders. The absence of abnormalities across key hepatic enzymes (ALT, AST), bilirubin fractions, and protein fractions supports a **normative** profile. The machine-learning model corroborates this assessment by flagging the dataset as **normal**.

Recommendations

- Maintain a balanced diet rich in fruits, vegetables, lean proteins, and whole grains to preserve hepatic function.
- Continue regular physical activity (moderate aerobic exercise, at least 150 minutes weekly).
- Hydrate adequately and limit alcohol intake to reduce hepatic strain.
- Schedule routine annual check-ups, including periodic liver function testing, particularly if lifestyle changes or

medication adjustments occur.

Disclaimer

This report provides general medical information and is not a substitute for individualized professional medical advice, diagnosis, or treatment. Consult a qualified healthcare provider for personalized guidance.